Software Requirements Specification - Addis Ababa Bus Tracking System

1. Introduction

This document states what the Addis Ababa Bus Tracking System must do, for the people who build, test and use it.

Purpose. Riders in Addis Ababa cannot see where a bus is or when it will arrive. The system shows live bus positions and arrival times on a phone, using the GPS of the bus crew's phone.

Scope. The system has three web apps and one backend: a rider app, a driver app, an admin app and a server with a database. It covers route numbers 49 and 61 today and is built to take more routes. It does not cover tickets, payment or fare information.

Definitions.

Term	Meaning
Route direction	One way of a route, for example 61 toward Legehar. It has its own road and ordered stops.
Shift (assignment)	A driver and a vehicle working one route direction.
Terminal	The first stop of a route direction, where buses wait.
ETA	The estimated time until a bus reaches a stop, shown as a range.
GPS point	One position reading with its time, sent by the driver's phone.
Fresh	A bus whose newest GPS point is no more than 90 seconds old.

2. Overall description

Product perspective. The driver's phone sends GPS points to the backend. The backend works out where each bus is, when it will reach each stop, and when buses leave the terminals. The rider app

reads those results. The admin app manages the fleet. Nothing depends on special hardware in the bus.

Users.

User	Goal	Login
Rider	See when the next bus comes and where it is	None
Driver (crew)	Share the bus position with little effort	Username and password
Admin	Manage drivers, vehicles and shifts and watch the fleet	Username and password

Operating environment. The apps run in a modern phone or desktop browser. The backend runs on Node.js with a PostgreSQL database that has the PostGIS extension. The rider app uses OpenStreetMap map tiles.

Assumptions and constraints.

- Each driver has a smartphone with GPS, a data connection and the browser allowed to use location.
- The driver app must stay open and unlocked while the bus runs. A browser can stop sending GPS in the background.
- Each route direction needs its road line and its ordered stops loaded into the database first.
- Estimates start from guessed stretch times and improve as real trips are recorded.
- The interface is in English. Stop names are also stored in Amharic.

3. Functional requirements

3.1 Rider app

ID	The rider app shall
FR-R1	Let a rider search stops and places by name, with at least 2 typed letters.
FR-R2	Show, for a chosen stop, the buses on the road with an arrival range in minutes, or "Arriving".

FR-R3	Show a reason instead of a time when a bus has no estimate, for example no recent signal.
FR-R4	Show buses that have not reached the road yet as upcoming, marked as a rough estimate.
FR-R5	At a terminal, show waiting buses, the bus that is boarding, the minutes since the last bus left and the usual gap.
FR-R6	Show how full each bus is (seats available, standing only, full) and whether it is standing still.
FR-R7	Let a rider pick a start place and a finish place and list the routes that connect them, each with the stop to wait at and the next bus.
FR-R8	Draw all routes of the trip on a map in different colours with their live buses.
FR-R9	Highlight one route and its bus when its card is tapped.
FR-R10	Show the results in a sheet that can be dragged to three heights: map only, half, and list only.
FR-R11	Keep the search, the chosen route and the map when the rider switches between the Trips and Stops tabs.
FR-R12	Remember the last five stops opened.
ID	The rider app shall
FR-R13	Show the rider's position on the map when the rider asks for it.
FR-R14	Refresh live data by itself and pause while the page is hidden.

3.2 Driver app

ID	The driver app shall
FR-D1	Let a driver log in and show the driver's shift for the day.
FR-D2	Let the driver choose the route direction and start the shift.
FR-D3	Send GPS points to the server in batches, each with its own time.
FR-D4	Keep points that cannot be sent and send them when the connection returns.
FR-D5	Send a point at least every 10 seconds while the bus stands still.
FR-D6	Let the driver set how full the bus is.
FR-D7	At a terminal, let the driver tap Departing now and Start return trip.
FR-D8	Let the driver end the shift.

3.3 Admin app

ID	The admin app shall
FR-A1	Let an admin log in with a separate admin account.
FR-A2	Show all running buses with their state and open warnings.
ID	The admin app shall
FR-A3	Let an admin add, edit and delete vehicles and drivers, and reset a driver's password.

FR-A4	Let an admin create a shift for a driver, vehicle and route direction, edit it, and end it.
FR-A5	Show the record of route changes made during shifts.

3.4 Backend

ID	The backend shall
FR-S1	Match each GPS point to the road of the bus's route direction and keep its progress along it.
FR-S2	Reject points that are too inaccurate or timestamped in the future.
FR-S3	Work out each bus's speed from its recent points and an arrival range for each stop ahead.
FR-S4	Treat a bus as having no recent signal when its newest point is older than 90 seconds.
FR-S5	Detect when a bus reaches a terminal and when it starts the other direction.
FR-S6	Raise warnings for wrong way, off the road, a wrong phone clock, no signal and overdue at a terminal, and hide wrong-way and off-road buses from riders.
FR-S7	End a shift automatically after 30 minutes without a GPS point.
FR-S8	Learn real stretch times every night and use them in place of guesses.
FR-S9	Accept requests to driver and admin functions only with a valid, unexpired token.

4. Non-functional requirements

ID	Area	Requirement
NFR-1	Timeliness	A GPS point sent from a bus shall show on the rider map within about 10 seconds when the connection is good.
NFR-2	Honesty	The system shall show a range and a reason instead of a guess when data is old or missing. It shall never show an exact time it cannot support.
NFR-3	Reliability	A lost connection shall not lose GPS points. They are queued on the phone and sent later.
NFR-4	Reliability	The server shall end dead shifts by itself, so buses that stopped sending do not stay on the map.
NFR-5	Usability	The rider app shall work on a phone screen with one hand. The main task, finding the next bus, shall take no more than two screens.
NFR-6	Usability	The crew shall be able to run a shift with few taps, without typing while driving.
NFR-7	Performance	A data refresh shall fetch only what the screen shows. The apps shall pause refreshing when the page is hidden.
NFR-8	Security	Passwords and login tokens shall be stored only as hashes. Logins shall expire after 14 hours.
NFR-9	Privacy	The system shall not collect any rider data. Only the bus crew's position is stored.
NFR-10	Portability	The apps shall run in current Chrome, Edge, Firefox and Safari on phone and desktop, over https.
NFR-12	Testability	Estimate, departure, trip and shift rules shall be pure functions with their own test scripts.

5. External interface requirements

User interfaces. Three mobile-first web apps. The rider app has a Trips tab (planner over a map, with a results sheet) and a Stops tab. The driver app has login, setup, shift and running screens. The admin app has Live, Vehicles, Drivers and Shifts pages.

Backend interface. A REST API over https that sends and receives JSON. Rider requests need no login. Driver and admin requests carry a token in the request header. The API allows requests from the web apps (CORS).

Hardware interface. The driver's phone GPS, read through the browser's location feature, with the position, accuracy and speed of each reading.

Software interfaces.

Interface	Used for
PostgreSQL with PostGIS	Storage, and distance and position along a road line
OpenStreetMap tile server	Map pictures in the rider app
Browser location and storage	GPS reading, and the queue of points not yet sent

6. Business rules and parameters

These limits are estimates until a real test ride measures them. They all live in one file, placeholders.ts

Rule	Value
A bus counts as fresh when its newest point is no older than	90 seconds
Standing still is shown after	2 minutes
An estimate that reads "Arriving" is	60 seconds or less

Rule	Value
Estimate range	plus or minus 20% (40% when the bus's own speed cannot be used)
Fallback speed when a stretch has no data	20 km/h

Upcoming buses are listed up to	60 minutes away
Usual wait at a terminal	10 minutes
Longest expected wait at a terminal	25 minutes
A bus is at a terminal within	100 m
A bus is off the road beyond	100 m
Admin no-signal warning after	180 seconds
A shift ends by itself after	30 minutes without a point
Server check for dead shifts	every 60 seconds
Crowding status counts as old after	10 minutes
A learned stretch time replaces a guess after	3 real crossings
Login token lifetime	14 hours